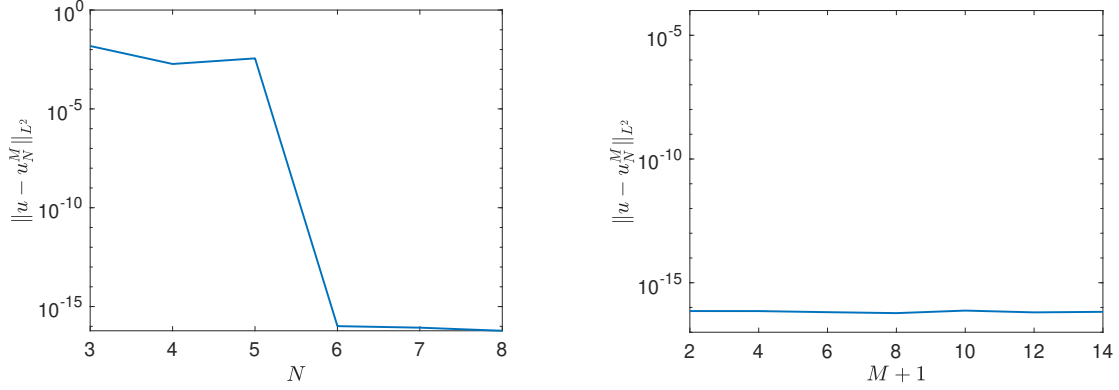


polynomial degree N at $M = 11$, while Figure 3(b) displays the L^2 -errors with respect to the angular discretization number $M + 1$ at $N = 30$.



(a) Numerical errors $\|u - u_N^M\|_{L^2}$ vs. N ($M = 11$). (b) Numerical errors $\|u - u_N^M\|_{L^2}$ vs. $M + 1$ ($N = 30$).

Figure 3: The L^2 -errors versus N and $M + 1$ of Example 1 using the spectral method.

From Figure 3(a), it is evident that the proposed method exhibits spectral accuracy concerning the polynomial degree N . As the degree N increases to 6, the L^2 -errors approach the machine epsilon, indicating the convergence of the numerical solution to the exact solution. Similarly, in Figure 3(b), it can be observed that when the angular discretization number $M + 1$ is small (e.g., $M = 1$), the L^2 -errors also approach the machine epsilon. This occurs because the exact solution is a sixth-degree polynomial solely in the spatial variable x and is entirely independent of the angular variable μ . As a result, selecting $N = 6$ and $M = 1$ enables the L^2 -error to reach machine epsilon.

Furthermore, by comparing Figure 2(b) and Figure 3(a), it is evident that for problems with a higher regularity of the exact solution, the spectral method requires fewer degrees of freedom compared to the Hermite WENO fast sweeping method to achieve the same level of accuracy.

5.2 Example 2

In this test, we solve the absorbing-scattering transfer problem described by the equation (1.1) with

$$\Sigma_t = 22000, \quad \Sigma_s = 1,$$

$$s(x, \mu) = -4\pi\mu^3 \cos^3 \pi x \sin \pi x + \Sigma_t(\mu^2 \cos^4 \pi x + \text{const}) - \Sigma_s(\text{const} + \frac{\cos^4 \pi x}{3}).$$

Here $\text{const} = 10^{-14}$ is a small positive constant which is used to ensure the source term to be nonnegative. The computational domain is $D = (0, 1)$. The boundary condition is given as follows

$$\begin{cases} \varphi(0, \mu) = \mu^2 + \text{const}, & \text{if } \mu > 0, \\ \varphi(1, \mu) = \mu^2 + \text{const}, & \text{if } \mu < 0. \end{cases}$$

For this problem, we have the exact solution given as [25]

$$\varphi(x, \mu) = \mu^2 \cos^4 \pi x + \text{const}.$$